

This assignment focuses on **Semantic Similarity**, which measures how closely two words are related in meaning. Unlike simple keyword matching, this looks at the "conceptual distance" between words.

In Natural Language Processing (NLP), this is often calculated using **Word Embeddings** (like Word2Vec or GloVe), where words are converted into vectors in a multi-dimensional space. The closer the vectors, the higher the similarity.

5 Word Pairs and Semantic Analysis

Here are five pairs ranging from high similarity (synonyms) to related concepts (associations).

Word Pair	Similarity Type	Explanation
Car & Automobile	Synonyms	These have near-identical semantic meaning. In a vector space, they would sit almost on top of each other.
King & Queen	Relational	They share the same "royalty" and "human" attributes but differ on the "gender" axis.
Coffee & Cup	Associative	They aren't the same thing, but they appear in the same context frequently, giving them a high co-occurrence score.
Doctor & Hospital	Functional	One is a professional, the other is the location. They are semantically linked through the domain of "Healthcare."

Happy & Joyful	Graduated	Both represent positive emotion, but "joyful" often implies a higher intensity than "happy."
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How Semantic Similarity is Measured

To explain this in your assignment, you should mention **Cosine Similarity**. This is the standard formula used to determine the angle between two word vectors.

The formula is:

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

- A score of **1** means the words are identical in context.
- A score of **0** means they are completely unrelated.

Simple Python Example

If you need to demonstrate this with code, you can use the Spacy library, which comes with pre-trained word vectors.

Python

```
import spacy
```

```
# Load a medium-sized model that includes word vectors
```

```
nlp = spacy.load("en_core_web_md")
```

```
word_pairs = [("car", "automobile"), ("coffee", "cup"), ("tree", "sandwich")]
```

```
for w1, w2 in word_pairs:
```

```
    token1 = nlp(w1)
```

```
    token2 = nlp(w2)
```

```
print(f"Similarity between {w1} and {w2}: {token1.similarity(token2):.2f}")
```

Would you like me to explain the difference between "Lexical Similarity" (how words are spelled) and "Semantic Similarity" (what words mean) for your report?